

Full length article

Fish avoidance of ships during acoustic surveys tested with quiet uncrewed surface vessels[☆]

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ABSTRACT

Acoustic surveys to estimate fish biomass and abundance are a major component of many fisheries monitoring programs. An important bias in acoustic surveys is that fish may avoid the survey vessel. Here, we utilized quiet uncrewed surface vessels (USVs) equipped with 120 kHz split beam echosounders to evaluate fish responses to motorized survey vessels. Two of these USVs were deployed in Lakes Michigan and Huron in summer 2021 and compared against three motorized vessels used in conventional fisheries acoustic surveys. Paired comparisons employed vessel-drone passes that provided the opportunity to observe fish response as vessels approached and then overtook their quiet USV counterparts. Sound originating from the vessels was primarily in the 10–1000 Hz range. Overall received sound pressure levels for ships at the closest pass were 100 dB (re 1 μ Pa @ 1 m) for one and 133–134 dB for the other two vessels. In contrast, sound originating from the USV was not detectable over ambient noise. We examined acoustic data from the USVs for potential changes in total acoustic backscatter, average target depth, and average in situ target strength as vessels approached. We observed weak evidence of an avoidance response by fish to the vessel with the loudest noise profile and highest survey speed but not for the other two vessels. We also compared acoustic data from 33 2-km transects surveyed by both vessels and the USVs, finding few differences between vessel and USV data for water depths between 5 and 80 m. Results from this work suggest that acoustics estimates of fishes in Lakes Michigan and Huron (primarily alewife, rainbow smelt, and bloater) are largely consistent among the vessels used in these two lakes for standard acoustic surveys and that fish avoidance is minimal in water depths > 5 meters.

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1. Introduction

Acoustic surveys are an important component of fish stock assessment programs for many of the world's largest fisheries, providing fisheries-independent data on spatially extensive fish populations (Rudstam et al., 2012; Simmonds and MacLennan, 2005). The accuracy of acoustic survey data depends on a range of factors, including the assumption that fish response to survey vessels does not change over time (ideally there is no response) and is consistent across vessels when more than one ship is used to assess the same fish stock (De Robertis and Handegard, 2013; Olsen, 1990). Fish avoid survey vessels either by diving and swimming away from the vessel (De Robertis and Handegard, 2013; Misund, 1990; Soria et al., 1996), changing shape of schools (Guillard et al., 2010), or potentially being herded in front of the vessel (Vabø et al., 2002).

Ship noise is believed to be the primary causal factor eliciting fish behavioral response to acoustic survey vessels (De Robertis et al., 2008; Mitson, 1995). Although, ship-based noise from motors, hydraulics, and propulsion can be of high intensity and propagate long distances (>10 km) through water (Merchant et al., 2014), avoidance may be limited to closer range to the ship. For example, avoidance by clupeids occurred at ranges of 100 s of meter (Vabø et al., 2002). Fish could also conceivably respond to visual cues from acoustic survey vessels (Brehmer et al., 2019; DuFour et al., 2018; Fréon and Misund, 1999); however, visual cues from approaching ships would likely be restricted to the upper water column (De Robertis and Handegard, 2013).

Detection of fish avoidance effects on acoustic surveys has been challenging to study (Knudsen et al., 2009; Olsen, 1990), and has produced variable results, suggesting vessel avoidance may be context-dependent. Several studies have reported vessel avoidance by

fish thereby biasing acoustic surveys low (De Robertis et al., 2010; DuFour et al., 2021; Elliott et al., 2021; Ona et al., 2007; Vabø et al., 2002). However, other studies have shown little or no effect of vessel on acoustic surveys (Draštík and Kubečka, 2005; Fernandes et al., 2000; Trumpickas et al., 2020) or even attraction to stationary or drifting vessels (Røstad et al., 2006). Although smaller vessels are believed to result in lower or little avoidance in lakes (Draštík and Kubečka, 2005), in at least one case, a larger (and presumably louder) vessel detected more fish than a smaller ship (DuFour et al., 2021). It is also unclear how avoidance behavior is associated with vessel noise. Presumably, noisier vessels cause greater fish avoidance (Mitson, 1995), but some studies have shown fish avoidance of quieted vessels and less bias using noisier vessels (Brehmer et al., 2019; De Robertis et al., 2008).

In the Laurentian Great Lakes, nighttime acoustic surveys are conducted annually with up to four vessels in each lake to estimate forage fish biomass and manage salmonid fisheries (Parker-Stetter et al., 2009). These ships likely vary in the amount of noise radiated and may therefore elicit different avoidance reactions from the fish surveyed. As a result, it is important for fisheries surveys to compare multiple ships across a large spatial scale and range of fish species. To address fish avoidance of surveying vessels, a multidisciplinary research team deployed 120-kHz split beam echosounders on three ships and two uncrewed surface vessels (USVs) powered by wind and solar in Lakes Michigan and Huron of the Laurentian Great Lakes (Fig. 1A). Due to its relatively small size and non-motorized propulsion, the USV provides a novel opportunity to evaluate fish response behavior to motorized survey vessels against an 'undisturbed' acoustic baseline. Three different scientific vessels, which routinely conduct annual fishery acoustic surveys in these lakes approached and overtook the USVs at night to evaluate if vessel proximity caused changes in acoustic scattering

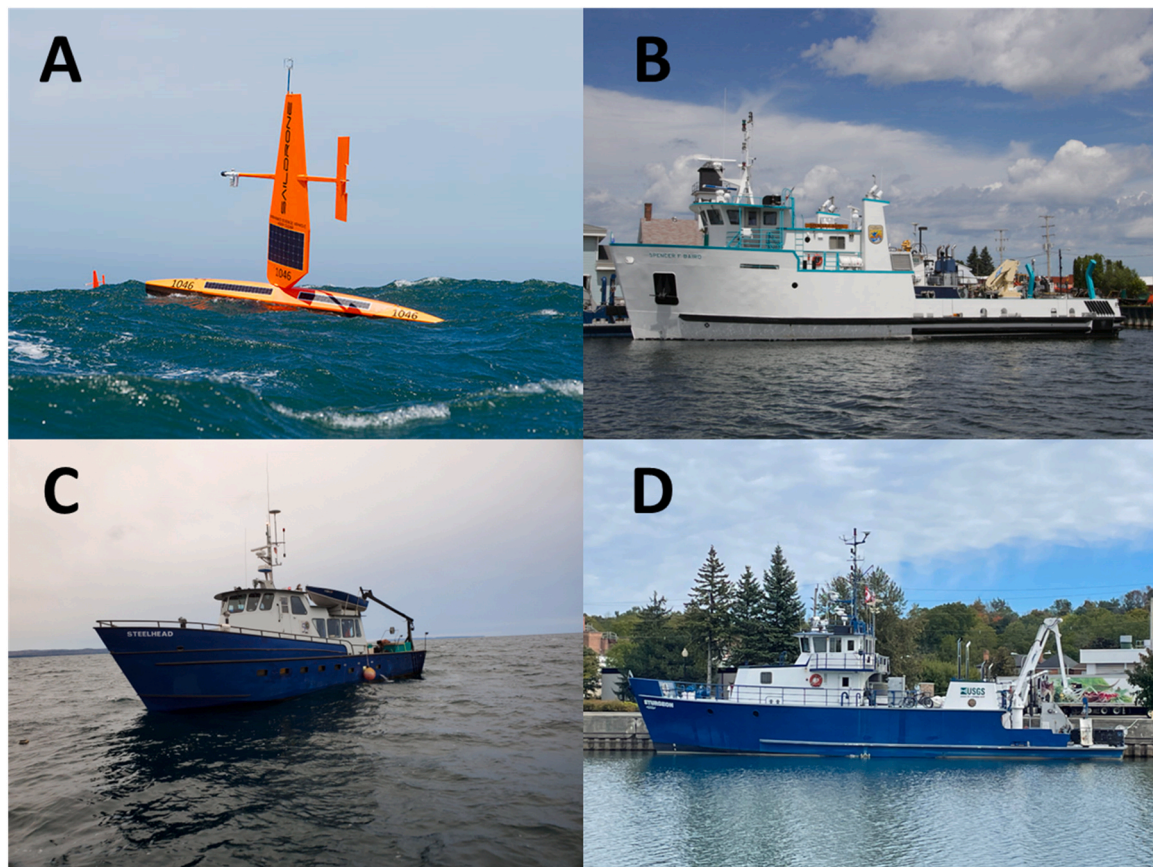


Fig. 1. Pictures of the A) Uncrewed Surface Vessel (USV, “Saildrone,” SD-1046; length = 7 m; image courtesy of Saildrone, Inc.), B) M/V Spencer F. Baird (length = 29 m), C) S/V Steelhead (length = 19.1 m), and D) R/V Sturgeon (length = 30.5 m) used to evaluate fish responses to motorized survey vessels conducting acoustic surveys in Lakes Huron and Michigan in summer 2021.

measured by the USVs. We hypothesized that as vessels approached the USVs, acoustic backscatter (measured as nautical area scattering coefficient, NASC), average target depth (TD), and target strength (TS) would decline and that this decline would be related to the amount of noise produced by the vessels. We expected to see the strongest response by the fish in the shallower epilimnion zone, less response from fish below the thermocline, and no response from fish at water column depths > 40 m based on diving observed in *Sardinella aurita* (Round sardinella), which was most pronounced in the first 20 m (Gerlotto and Fr  on, 1992). To test these hypotheses and evaluate fish responses to motorized survey vessels on the Great Lakes, our objectives were to: 1) characterize noise levels of sound emanating from three motorized survey vessels and the USVs and compare them against acoustic vessel sound standards suggested by the International Council for Exploration of the Sea (ICES) (Mitson, 1995), 2) assess fish responses from USV acoustic measurements as motorized vessels approached, and 3) compare standard acoustic survey metrics of target depth, target strength, and NASC levels between motorized vessels and quiet USVs.

2. Materials and methods

2.1. Vessel acoustic profiling

Three ships (M/V Spencer F. Baird, S/V Steelhead, R/V Sturgeon, hereafter referred to as Baird, Steelhead and Sturgeon, respectively) and two Uncrewed Surface Vessels (USVs, "Saildrones" from Saildrone Inc.) were used (Table 1; Fig. 1). The noise profiles from each motorized vessel and one USV were each characterized using an underwater array of hydrophones (Fig. S1). The underwater array consisted of three hydrophones (Teledyne RESON TC4032), each spaced 1 m apart, anchored on the lakebed at bathymetric depths of ~30–40 m. The underwater hydrophone array was tethered to the data acquisition vessel located ~40–50 m away. A line of marking buoys ran from overtop the underwater array to the measurement vessel to designate the course for the vessel being profiled. The vessel would then run the course 1 km to either side of the measurement location (2 km total). Each run was recorded using a handheld data recorder (Crystal Instruments CoCo-80X). GPS data were also collected for the location of the underwater array, measurement vessel, and the path of the scientific vessel during profiling.

The vessels were profiled while traveling in their acoustic survey operating conditions (i.e., standard operating speed and generally with

hydraulics off [if present]; Table 1). Each vessel made two runs of the test course, in opposite directions, and the sound levels were averaged for the same locations, relative to the hydrophone array, on the two runs. The hydrophone measurements from the underwater array were filtered between 11 Hz and 5680 Hz, and then beamformed using traditional delay-and-sum beamforming and the physical location of the array and vessel. The resulting acoustic spectra were then analyzed at two points along the course: 200 m away from the underwater array as the vessel was approaching and the closest point the vessel was to the underwater array.

Sound speed profiles were recorded at each measurement location. The mean sound speeds were similar for all survey vessels: 1476 m•s⁻¹, 1468 m•s⁻¹, 1472 m•s⁻¹, and 1481 m•s⁻¹ for the Baird, Steelhead, Sturgeon, and USV, respectively. The sound speed profiles were all monotonically decreasing with depth indicating a slight downward refracting condition. Variations in sound speed from top to bottom of the water column were 0.2%, 3.3%, 1.6%, and 2.3% for the Baird, Steelhead, Sturgeon, and USV, respectively. Because of the relatively small change in sound speed with depth, combined with the close measurement range, low frequency of the ship noise, and variability in background at test sites, corrections for the acoustic channel were not done and received levels are reported here for comparison. Each measurement was averaged for 10 s during which the vessels were moving, which helped to mitigate, but not eliminate, the acoustic channel effects through averaging. The acoustic signatures of the Sturgeon was compared with the underwater radiated noise specification proposed by Mitson (1995).

For comparisons of overall platform noise, the acoustic profile was integrated over the range of measured values (10–6000 Hz) and an overall, received sound pressure level (in dB; re 1 µPa @ 1 m) and received octave band sound pressure levels were calculated.

2.2. Uncrewed surface vessel

We used two USVs ~7 m in length powered by wind and solar and constructed of carbon fiber (Fig. 1A). Propulsion is provided by a 4-m sail instead of a propeller and therefore these USVs are substantially quieter than traditional ships. Solar panels were incorporated into the sail and body of the drone to provide power for onboard devices including actuators for vehicle control, communication devices, and scientific equipment. At night a white light on top of the USV was turned on to help other vessels avoid collisions. For this study, two of these drones were each equipped with a split beam 120 kHz transducer

Table 1

Vessel and acoustic receiver information used to evaluate fish responses to motorized survey vessels in Lakes Huron and Michigan in summer 2021.

	M/V Spencer F. Baird	S/V Steelhead	R/V Sturgeon	SD-1032	SD-1046
Type	Motorized vessel	Motorized vessel	Motorized vessel	"Saildrone"	"Saildrone"
Length (m)	29.0	19.1	30.5	7.0	7.0
Beam (m)	9.1	5.0	7.8	1.0	1.0
Draft (m)	2.9	2.0	3.0	2.0	2.0
Displacement (tonnes)	232	63.5	182	0.5	0.5
Number of engines	2	2	2	0	0
Number of blades on prop	4	4	3	0	0
Shaft RPM during survey	1200	1300	750	NA	NA
Survey speed (m/s) (mean ± SD)	4.5 (± 0.1)	4.3 (± 0.9)	2.0 (± 0.3)	1.0 (± 0.4)	0.7 (± 0.3)
Hydraulics off during acoustic survey	Yes	Yes	No	NA	NA
Mean 200 m Approach Noise Level dB re 1µPa (variation between runs – dB)	117 (11)	98 (4)	126 (28)	NA	NA
Mean Closest Pass Noise Level dB re 1µPa (variation between runs – dB)	133 (14)	100 (0)	134 (6)	NA	NA
Acoustic Unit	Simrad EK80	BioSonics DTX	BioSonics DTX	Simrad EK80	Simrad EK80
Transducer mounting	Moon pool	Pole mounted	Moon pool	Keel	Keel
Transducer Depth (m)	3.00	1.30	2.05	2.00	2.00
Frequency (kHz)	120	123	123	120	120
Source level (dB re 1µPa)	209	221	220	209	209
Pulse Duration (ms)	0.256	0.400	0.400	0.256	0.256
Background S _v noise dB re 1 m ⁻¹ @ 1 m	-130	-119	-120	-116 (max: -108)	-117 (max: -111)

(Simrad EK80) with a beam angle of 7°. The USV communicated via satellite link every 10 min, allowing an operator to provide waypoints and set courses remotely. Additionally, the transducer's ping rate, pulse length, and maximum depth of data collection could be altered by the operator at those times. However, the USVs were not able to transmit acoustic data due to the large data volume involved; instead, all acoustic data were stored onboard internal hard drives until the completion of the mission. Meinig et al. (2019) provides a full description of the USVs deployed in the current study.

The transducers mounted on the USVs were calibrated with a 38.1 mm tungsten carbide sphere before deployment dockside in Alameda, California (San Francisco Bay), and again post-deployment in Saginaw Bay, Michigan (Lake Huron). The calibration sphere was at 7 m depth with data analyzed with the calibration assistant (Echoview Software Pty Ltd, 2021). The more limited post-survey calibrations were conducted to check the stability of the sounders. The measured on-axis TS of the sphere was within ~ 0.2 dB re 1 m^2 of the value obtained in the pre survey calibration (Table S1). Here, we used the calibration values from the pre-survey measurements (Table S1).

Both USVs had an issue with electrical interference from a rudder actuator used to navigate the drone, and as a result background noise was higher for the drones than the powered vessels (this design problem was subsequently corrected for future USV deployments after the completion of this study). While the transducer used in the USV is known to operate with background noise levels lower than -140 dB re 1 m^{-1} (S_v) at 1 m, during our deployments the noise levels were about -115 dB re 1 m^{-1} at 1 m in both drones, at times as high as -108 dB re 1 m^{-1} at 1 m (Table 1). Background noise levels for the ship-based units were lower, -119 to -130 dB re 1 m^{-1} at 1 m (Table 1). Note that high “background noise” in the USV data is not related to the noise from engines radiated into the water column discussed above but the result of electrical interference in the cables from the transducer to the recording unit. This electrical interference limited the utility of acoustic data from deeper water to targets > -50 dB re 1 m^2 . Smaller fish with a TS of -60 dB re 1 m^2 could be measured without bias only down to 40 m depth with these background noise levels (Parker-Stetter et al., 2009; Rudstam et al., 2009). As a result, for all vessels, we investigated echoes from fish with TS > -60 dB re 1 m^2 in 5–40 m deep water, and echoes from fish with TS > -50 dB re 1 m^2 in the hypolimnion (40–80 m depth). Targets of -50 dB re 1 m^2 were detectable without bias at 80 m with background noise levels of -110 dB re 1 m^{-1} at 1 m when using these 7-degree transducers (see discussion in Rudstam et al., 2009).

2.3. Survey vessel-drone overtakes

To enable comparison of acoustics signals across platforms, the three survey ships approached and passed the USVs at close range (usually <100 m) on different nights (pelagic fish surveys in the Great Lakes are conducted at night) and in different locations (Fig. 2). Care was taken to operate the vessels at the same speed and illumination as during standard acoustic surveys (Table 1). These vessels were much larger than the USVs and differed in terms of shape, noise, and survey speed (Table 1). The Baird used a 120 kHz transducer (Simrad EK80); the Steelhead and Sturgeon used 123 kHz transducer (BioSonics DTX units; Table 1). The Steelhead and Sturgeon used longer pulse durations than the Baird and the drones (0.400 ms vs. 0.256 ms; Table 1). Pulse duration affects the resolution of individual targets but this was not a major issue here as fish density was low and fish were dispersed at night (for an example echograms see Fig. S2). The ships acoustic units were calibrated with tungsten carbide standard spheres (38.1 mm for Baird and Sturgeon, 33.2 mm for Steelhead). The Baird was calibrated August 7 in Lake Michigan and September 30 in Lake Huron, the Sturgeon on August 8 in Lake Michigan, and September 9 in Lake Huron, while the Steelhead was calibrated on July 15 (Table 1). The post-survey calibrations for the Baird and the Sturgeon were consistent with the pre-survey calibration (within ~ 0.1 dB of sphere TS) indicating stable performance of these units (Table S1) so we used the pre-survey values in our analyses. The Steelhead unit was only calibrated pre survey (Table S1).

The USV moves considerably slower than the ships at standard survey speed (Table 1), so we could not use the standard vessel comparison design of one ship following another at some constant distance. Instead, we used an ‘overtake’ approach, where the USV was initially ahead of the vessel with the vessel catching up to the drone. Overtakes were always conducted at night between 2200 and 0300 (Eastern Daylight Time, EDT; Fig. 2). Ships attempted to approach the USV starting ~ 2 km astern, following the track of the USV and leaving the USV track after the overtake, avoiding passing in front of the USV as they turned to avoid including bubbles from the wake of the vessels in the overtake data. We used 2 km transects to provide sufficient starting distance between the two platforms for which vessel noise was assumed to have no impact on fish behavior based on work in marine environments which detected fish responses at 100's of meters (Vabø et al., 2002). Several overtakes were completed on the same night by allowing the USV to survey the water first and the ship following the USV after it had moved ~ 2 km further along its preset course; in this manner, up to four overtakes were

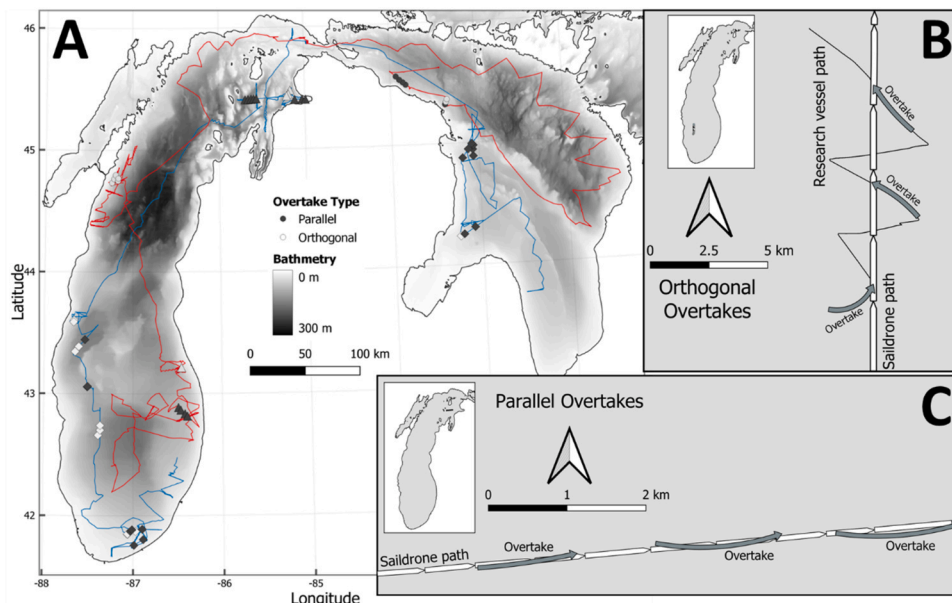


Fig. 2. A) Locations where overtakes (i.e., transects where drone and vessel survey the same water at the same time and the vessel approached the drone from astern) of Uncrewed Surface Vessel (USV) by one of three ships occurred in Lakes Michigan and Huron, with the colored paths taken by each USV (red for SD-1032 and blue for SD-1046) over the course of the study. The shape of the symbols is the vessel overtaking the uncrewed surface vessel (circle for M/V Spencer F. Baird, triangle for S/V Steelhead, and diamond for R/V Sturgeon). The type of overtake could be B) either orthogonal or C) parallel (see Section 2.3 for details).

collected on the same night. The time from the start of the overtake 2 km from the USV to the ship passing the USV varied from 8 to 15 min depending on the speed of the ship and the variable speed of the USV. Due to the practicalities of ship navigation some overtakes approached the USV from an angle and therefore did not survey the same swath of water. We call these overtakes “orthogonal,” although the angle of approach varied. Examples of a parallel and orthogonal overtake are provided (Fig. 2).

We used two different analyses for the overtakes. First, we used both the parallel and the orthogonal overtakes to examine potential fish responses in USV acoustics signals as vessels approached. This is similar to the method used by Vabø et al. (2002) and others with a survey vessel passing over a moored buoy with an acoustic transducer (De Robertis and Wilson, 2010; Handegard, 2003). Second, we directly compared USV to the ship acoustic data integrated over 2 km parallel tracks. This is similar to the approach of a vessel following another for ship-to-ship comparisons often used in marine surveys (De Robertis et al., 2008) and employed before in the Great Lakes (DuFour et al., 2021). In one case, an overtake began as orthogonal and became parallel halfway (~1 km) through (16 September 2021 with the Sturgeon); only the parallel section of this overtake was considered parallel.

2.4. Acoustic analysis

The lake bottom was identified using the best bottom candidate function in the acoustic processing software (v12.1 Echoview), and then reviewed and manually edited. A 0.5 m back-step was added to the identified bottom to exclude any bottom effects in the analysis. If the bottom was below the maximum depth used for data storage, the lower limit was set to 80 m. All platforms had transducers located between 1.3 and 3 m depth (Table 1); to ensure that all vessels compared the same water, we excluded all data within 5 m of the surface (the depth of the Sturgeon transducer plus 2 m nearfield). We manually reviewed the excluded surface zone to ensure that no backscattering from wave induced bubbles was included in the analysis. When the vessel and USV were within ~50 m of each other, we detected crosstalk between the ship and the USV sounders (Fig. S2) and we removed sound spikes in those areas. We also applied a background noise removal filter in the acoustic processing software (v12.1 Echoview; horizontal extent: 20 pings, vertical extent: 5 samples, maximum noise: -110 dB, based on De Robertis and Higginbottom, 2007) to S_v data.

Target strength thresholds were set to detect echoes from targets larger than -60 dB re 1 m² in water shallower than 40 m (epilimnion and metalimnion) and from targets larger than -50 dB for water 40–80 m (hypolimnion). A minimum TS of -60 dB re 1 m² is commonly used in Great Lakes surveys, but most hypolimnetic fish are larger with TS greater than -50 dB re 1 m² in the Great Lakes (Parker-Stetter et al., 2009; Rudstam et al., 2009). Fish targets were well separated in the water column (Fig. S2) during all overtakes. After thresholding, single targets from the TS data were identified using Echoview’s “single target detection method 2 for split beams”. The defaults used for single target detection were the same for all units (default values: maximum beam compensation: 4 dB, standard variation for minor and major axis angles 0.6, echo lengths between 0.6 and 1.5 times the pulse length). We used a maximum beam compensation of 4 dB, that is lower than the recommended value (6 dB) in the Great Lakes standard operating procedure (Parker-Stetter et al., 2009). This allowed unbiased detection of smaller targets in somewhat deeper water than with the recommended 6 dB (see Rudstam et al., 2009).

To limit backscattering from non-fish targets, we applied a “Minimum TS threshold” to the S_v data. This threshold was set to 6 dB below the minimum target strength of interest (equates to -66 dB re 1 m² and for depths of 40–80 m and -56 dB re 1 m² for deeper waters). The “Minimum TS threshold” option for echo integration is often used in freshwater applications to remove backscattering from smaller targets like zooplankton and mysids. The “Minimum TS threshold” is set to 6 dB

below the smallest target of interest to pick up backscattering from that target when located at the edge of the beam and is then transformed to an actual time-varying S_v threshold based on the differences in the Time Varied Gain (TVG) functions for S_v and TS. See Rudstam et al. (2009) for more discussions on using that threshold approach. The S_v data and single-target data were divided into 1.2 s by 1 m depth cells and exported to R for further processing (version 4.1.3; R Core Team, 2023). In R, the mean nautical area backscattering coefficient (NASC) was calculated from the S_v data by summing the acoustic backscatter over the depth of interest and multiplying by $4\pi(1852)^2$, and the mean target depth and target strength from the detected single fish targets for each overtake.

For analysis, we divided the water column into three layers: epilimnion (5–17 m), metalimnion (18–40 m), and hypolimnion (41–80 m). These depth layers aligned with temperature profiles that showed a thermocline at ~15–20 m, and temperatures of ~7.5 °C at 40 m in Lakes Michigan and Huron during the 2021 summer survey period (Fig. S3). Annual fisheries surveys have demonstrated that depth layers have different fish species/sizes (Fig. S4; Warner et al., 2012), and each depth layer in our study showed distinct TS distributions indicating different fish assemblages. Fish present in each depth layer were sampled with depth-targeted midwater trawls that are part of the standard ship-based surveys to determine species compositions (Warner et al., 2022). Here we present the dominant fish species in each layer inferred from the trawl data (details in Hondorp et al., 2022 and Warner et al., 2022).

2.5. Analysis of USV data for potential fish responses to approaching vessels

To evaluate whether fish changed their behavior in response to an approaching ship, the acoustic response measured by the USV was regressed against the distance of the vessel from the USV. For this analysis we evaluated all vessel-drone overtakes (parallel and orthogonal; $n = 55$). We considered data collected from the time a vessel was 2 km from a USV until the vessel overtook the USV. We divided the USV data collected up to the time of the ship passing the USV into 30 s intervals. We considered using smaller intervals but reducing interval size resulted in numerous intervals with few targets and increased variability in the response. A 30 s interval was the smallest interval we were comfortable using due to the low fish density in many transects. As a result, our approach may have been too coarse to detect small differences between USV and vessel. The calculated distance from the vessel to the USV at the midpoint of each interval was then used to estimate changes in USV acoustic response as the vessel approached. The last interval was not always a full 30 s, as a result we dropped any interval that was shorter than 15 s. Each 30 s interval was also separated into three depth layers (epilimnion [5–17 m], metalimnion [18–40 m], hypolimnion [41–80 m]). The average NASC, target depth (TD), and TS were calculated for each remaining interval and depth region using values from the original 1.2 s by 1 m cells exported from Echoview.

We fit Generalized Additive Models (GAM) to USV data to allow for nonlinear signatures of acoustic data reflecting potential fish responses as vessels approached. For example, prior studies have suggested that the zone of influence of motorized survey vessels on fish behavior may be on the order of 200 m or less and areas farther away may be effectively undisturbed (De Robertis et al., 2019). Thus, we expected any fish behavior changes reflected in acoustic data would manifest as a response when ships passed within some distance less than 300 m from the quiet drones.

Generalized Additive Models were implemented in the mgcv R package (version 1.8–40; Wood, 2012) and fit separately for each vessel, lake, and depth layer separately with the following formula:

$$\text{Response} = \alpha_0 + s(D) + \epsilon,$$

where α_0 is the intercept, $s()$ is a smoothing function using a cubic spline basis, D is the distance in negative km from the USV to the vessel (i.e., -2 km would be the approximate start point) at the midpoint of the interval, and ε is a normally distributed error term. Response variables were NASC, TD, or TS. Values varied substantially across overtakes within a given response variable. Thus, we rescaled response data into Z-scores at the level of each overtake (i.e., combined across all 30 s bins within a given overtake) to standardize values across separate regressions. Z-scores convert values in a sample into units of standard deviations from the sample mean [$Z\text{-score} = (\text{data value} - \text{sample mean}) / \text{sample standard deviation}$] and can facilitate comparison of data across different scales under a common currency. General Additive Models were fit with three knots to account for potential non-linear effects while keeping the complexity of fitted regression relationships parsimonious; GAMs fit with five knots were also explored however results were insensitive to the choice of knots. General Additive Model fits indicating a flat line with a zero intercept suggest a lack of fish response to approaching vessels, while a positive slope or negative slope suggests avoidance.

2.6. Comparison of USV and vessel estimates integrated across 2 km parallel overtakes

Parallel overtakes ($n = 33$) provided an opportunity to directly compare acoustic signal from motorized vessel and quiet USVs integrated over the entirety of the parallel transects (~ 2 km of water). Because both platforms surveyed the same water within ~ 20 min windows, we assumed the same fish were surveyed by both vessel and USV

and therefore that the differences in these 2 km sections for NASC, S_A , TD or TS were related to fish response to the vessels. For example, fish avoidance behavior in response to vessels should result in lower backscatter if fish dive (change orientation) or swim out of the path of the vessel's acoustic beam. We therefore expected NASC and TS to be lower and TD deeper in the vessel data compared to the USV data. The S_A (nautical area scattering strength) is in dB re $1 \text{ (m}^2 \bullet \text{nmi}^{-2})$ and the difference in S_A values therefore represents the ratio of the NASC values ($\Delta S_A = 10 \bullet \log_{10}[\text{NASC}_{\text{SD}}/\text{NASC}_{\text{VES}}]$).

Further, we expected the difference between USV data and vessel data to be related to vessel noise levels. Each parallel overtake was divided into the depth layers described above (e.g., epilimnion, metalimnion, hypolimnion) and the response averaged across all cells in a layer. The difference between USV and ship for each depth layer was calculated for NASC, TD, and TS and compared using Analysis of Variances (ANOVA) tests. We could not include lake in these models because only the Sturgeon made parallel overtakes in both lakes. We used the 95% and 99% confidence intervals to evaluate whether the difference between USV and vessel was different from zero. We only used layers from transects where the USV detected at least 10 targets to minimize the effect of variability due to small sample sizes.

3. Results

3.1. Vessel noise

Vessel received sound pressure levels grew louder as vessels approached the listening array (Table 1; Fig. 3, Figs. S1, S5; sea state:

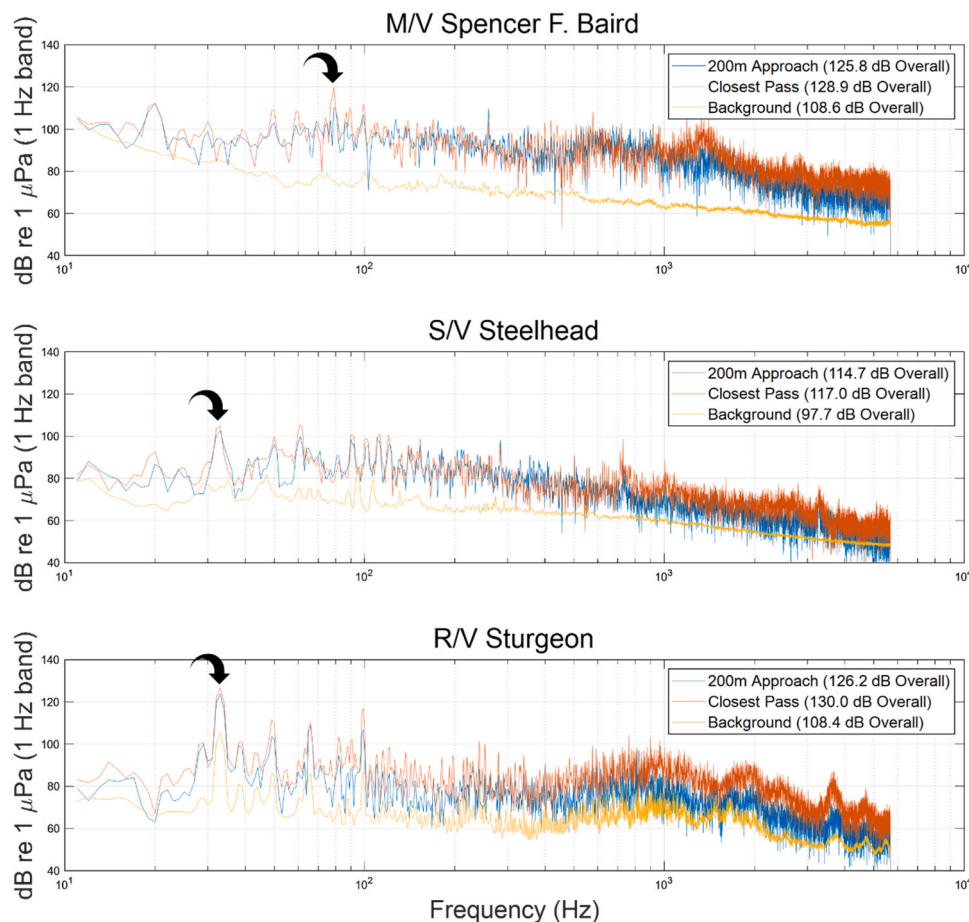


Fig. 3. Comparison of the received acoustic profiles for each of the three scientific survey vessels at 200 m approaching the underwater array compared to the closest pass to the array. The black arrows are the blade pass frequency. Background measurements are shown for comparison. Note that the background measurement for the R/V Sturgeon captured some of the vessel's operating noise in the distance as it transited away from the test site.

~0). The USV signature was below the ambient noise levels of the surface-generated noise on the day of the test (sea state: ~4–5). Because there was no measurable change in sound pressure levels between the background measurement and the USV measurement, it can be assumed that the USV signature is 10 dB or more below the ambient levels on this test day (Fig. S6). The Baird was loudest and had an averaged overall received sound pressure level of 117 dB (re 1 μ Pa at 1 m) at the 200 m approach and 133 dB at the closest pass. The Sturgeon had an averaged overall received sound pressure levels of 126 dB at the 200 m approach and 134 dB at the closest pass. Lastly, the Steelhead was the smallest and quietest motorized vessel and had an averaged overall received sound pressure levels of 98 dB at the 200 m approach and 100 dB at the closest pass.

There was significant variability between the two runs for some vessels (see Table 1). The relatively large variation between the 200 m approach runs for the Sturgeon, 28 dB, is unexplained. Whereas the Steelhead varied by less than 4 dB between runs, although it was measured at a different test location. The variation in the closest pass received sound pressure levels is assumed to be due to the vessel not passing directly over the hydrophone array, creating slight angle of arrival differences from run-to-run.

The range-corrected sound pressure level spectra of the Sturgeon matched the shape of the Mitson (1995) curve, including the knee frequency and slope of the high frequency roll off at ~1000 Hz (Fig. S7). The Sturgeon range-corrected values are shown using both spherical and cylindrical spreading (Fig. S7); in the shallow channel where these measurements were recorded, the true transmission loss is more complicated and likely lies somewhere between these lines. No noise control treatments have been implemented on these three vessels, and as a result high amplitude, low frequency peaks that rise 10–45 dB above the ICES recommendation are unsurprising. The peaks in received sound pressure levels at low frequencies (e.g., the ~31 Hz peak in the Sturgeon and Steelhead data and the ~20 Hz peak in the Baird data; Fig. 3, Fig. S5) and their harmonics were due to the blade pass frequency (number of blades multiplied by the main shaft rotations per minute) of each vessel. As a result, this octave band often had the highest, or close to the highest, received sound pressure level (Fig. S5). At frequencies greater than 200 Hz, the noise becomes more broadband and is a result of radiated noise from many mechanical sources on the vessel as well as flow noise from the propeller.

3.2. USV flight characteristics

Mean windspeeds for nights on which overtakes occurred were $< 8 \text{ m}\cdot\text{s}^{-1}$ (Table S2), with an overall mean windspeed during overtake tracks of $4.8 \text{ m}\cdot\text{s}^{-1}$. USV pitch (average = 1.22° , SD = 0.07°) and roll (average = 0.19° , SD = 1.85°) during overtakes was limited and we did not observe pings indicating poor data quality.

3.3. All overtakes: Fish behavioral response

We observed a limited response of fish to approaching ships when examining USV data from parallel and orthogonal overtakes (Table 2, Figs. 4–6). Most GAMs (36 of 45 models) did not show support for a statistically significant smooth term, i.e., indicating that the fitted regression curves were not supported as being different from a zero-slope line relationship on the distance between USV and ships for a given response variable and thus were reflective of a lack of fish response to approaching vessels (Table 2; Figs. 4–6). Note that based on the number of models (45), approximately two would be expected to be significant by chance alone under a null hypothesis of no difference between acoustic platforms with an alpha level of 0.05. Seven of the nine GAMs with statistically significant smooth terms produced a linear fit (EDF = 1) or nearly so (EDF < 1.1), whereas the remaining models with a statistically significant smooth term had an EDF ≤ 1.9 and indicated a nearly linear fit (Table 2). Contrary to our hypothesis that the effect

would be greatest in the epilimnion, statistical support for an effect of an approaching vessel was equally common in the hypolimnion and the epilimnion (each had four significant models), while one was significant in the metalimnion (Table 2). Seven of the models supporting a fish response were associated with the Baird and two were associated with the Steelhead. None was associated with the Sturgeon (Table 2).

The direction of the effect was also variable. Four of the significant models had NASC as the response (4 out of 15 NASC models) and there was a significant increase in NASC in three models (epilimnion and metalimnion for the Baird in Michigan and hypolimnion for the Steelhead in Michigan) suggestive of fish accumulating in front of a moving vessel and then passing the USV just before the vessel overtook the drone. There was no apparent pattern to suggest the species composition played a role in these effects. Three models were significant for TD and two for TS (out of 15 modes for each variable). Two significant TD models indicated deeper fish distributions as the vessel approached suggesting fish move deeper as a response to the survey vessels (both in the Baird), but in one (the epilimnion with the Steelhead) fish depth increased as the vessel approached. However, in only one case where depth declined (Baird hypolimnion of Lake Huron) did we observe the expected decline in TS of diving fish. One TS model had an increasing TS (epilimnion – Baird in Michigan) and one had a decreasing TS (hypolimnion – Baird in Huron) as the powered vessel approached.

3.4. Parallel overtakes: Differences between USV and vessel

During the parallel overtakes, the same fish population was surveyed first by the USV and within a ~20-minute window also by the survey vessels. These two transects are therefore directly comparable and survey as close to the same fish population as possible with these vessels and USVs. Parallel overtakes occurred with the Baird (5 transects in Lake Huron), the Sturgeon (8 transects in Lake Huron and 8 in Lake Michigan), and the Steelhead (12 transects in Lake Michigan).

Sampling variability differed across ships and acoustic metrics. Power to detect statistically significant differences between USV and ships was variable with relatively greater precision in minimum detectable differences for S_A , TD, and TS compared to NASC values (Table S3). Combined across trials, however, statistical support for differences in acoustic data between USVs and vessels was weak and variable across lakes, vessels, and water column zones. The only statistically significant results were higher NASC and S_A in the metalimnion for Baird data and higher NASC and S_A in the hypolimnion for Sturgeon data (Fig. 7). Given the number of models (36 ANOVAs), we anticipated that one or two would be significant by chance alone under the null hypothesis of no difference between acoustic platforms. These two comparisons showed higher NASC and S_A for the USV than a vessel. For the epilimnion where we expected the largest difference, USV data was not higher than that obtained for any vessel. For contrasts with significant NASC differences, the response magnitude ranged from 3.4 to $5.8 \text{ m}^2\cdot\text{nmi}^{-2}$ and S_A values which ranged from 1.4 to 4.4 dB re 1 ($\text{m}^2\cdot\text{nmi}^{-2}$) (Fig. 7). There were no differences between USV and any ship for mean TD or TS (Fig. 7). There were also no differences in effects among vessels (Fig. 7). Given variability in the significant effects for NASC and the lack of response in TD or TS, these results indicate no systematic difference between ship-based data versus the USV acoustic data.

4. Discussion

Contrary to our expectations we detected few behavioral responses of fish to three different motorized survey vessels in our study. The loudest survey vessel (Baird) did elicit a possible fish behavioral response most often, but relatively few overtakes contributed to the analysis of USV data for this ship (Table 2; Figs. 4–6). Furthermore, the type of fish response indicated by regression models was inconsistent across transects (Figs. 4–6), suggesting that fish spatial variability, and

Table 2

General additive model (GAM) regression results for potential fish response signatures in Uncrewed Surface Vessels (USVs) acoustic measurements as each of three ships approached for each depth layer, vessel, and lake, across three acoustic metrics. The number of overtakes which contributed, and the number of 30-sec intervals (n) which were used to fit the line, are also reported. The suspected fish species are the predominant species found by annual prey fish trawl surveys in these layers, and the age-class(es) (YOY = Young-of-year, YAO = Yearling and older), if known from the trawl surveys, are also reported.

Depth layer	Vessel	Lake	Response	Overtakes	n	EDF [†]	F-value	R ²	Suspected species [‡]
Epilimnion	Baird	Lake Huron	NASC	7	145	1.52	2.11	0.01	Various species
Epilimnion	Baird	Lake Michigan	NASC	5	94	1.39	4.14 *	0.08	Alewife (<i>Alosa pseudoharengus</i>)
Epilimnion	Steelhead	Lake Michigan	NASC	15	341	1	0.15	0.00	Alewife (<i>Alosa pseudoharengus</i>)
Epilimnion	Sturgeon	Lake Huron	NASC	9	560	1	0.14	0.00	Various species
Epilimnion	Sturgeon	Lake Michigan	NASC	17	734	1	0.18	0.00	Alewife (<i>Alosa pseudoharengus</i>)
Epilimnion	Baird	Lake Huron	TD	7	77	1	0.45	-0.01	Various species
Epilimnion	Baird	Lake Michigan	TD	5	20	1	10.87 **	0.34	Alewife (<i>Alosa pseudoharengus</i>)
Epilimnion	Steelhead	Lake Michigan	TD	15	147	1	8.30 **	0.05	Alewife (<i>Alosa pseudoharengus</i>)
Epilimnion	Sturgeon	Lake Huron	TD	9	267	1.81	2.24	0.02	Various species
Epilimnion	Sturgeon	Lake Michigan	TD	17	530	1	1.22	0.00	Alewife (<i>Alosa pseudoharengus</i>)
Epilimnion	Baird	Lake Huron	TS	7	77	1	0.72	0.00	Various species
Epilimnion	Baird	Lake Michigan	TS	5	20	1	10.47 **	0.33	Alewife (<i>Alosa pseudoharengus</i>)
Epilimnion	Steelhead	Lake Michigan	TS	15	147	1.64	2.48	0.02	Alewife (<i>Alosa pseudoharengus</i>)
Epilimnion	Sturgeon	Lake Huron	TS	9	267	1	1.34	0.00	Various species
Epilimnion	Sturgeon	Lake Michigan	TS	17	530	1.66	0.90	0.00	Mostly YOY bloater (<i>Coregonus hoyi</i>), alewife (<i>Alosa pseudoharengus</i>)
Metalimnion	Baird	Lake Huron	NASC	7	145	1.75	1.47	0.02	Rainbow smelt (<i>Osmerus mordax</i>)
Metalimnion	Baird	Lake Michigan	NASC	5	94	1.07	8.43 **	0.09	Mostly YOY bloater (<i>Coregonus hoyi</i>), alewife (<i>Alosa pseudoharengus</i>)
Metalimnion	Steelhead	Lake Michigan	NASC	16	341	1	0.43	0.00	Mostly YOY bloater (<i>Coregonus hoyi</i>), alewife (<i>Alosa pseudoharengus</i>)
Metalimnion	Sturgeon	Lake Huron	NASC	9	560	1.04	0.30	0.00	Rainbow smelt (<i>Osmerus mordax</i>)
Metalimnion	Sturgeon	Lake Michigan	NASC	17	734	1	0.98	0.00	Mostly YOY bloater (<i>Coregonus hoyi</i>), alewife (<i>Alosa pseudoharengus</i>)
Metalimnion	Baird	Lake Huron	TD	7	115	1	0.71	0.00	Rainbow smelt (<i>Osmerus mordax</i>)
Metalimnion	Baird	Lake Michigan	TD	5	47	1	3.34	0.05	Mostly YOY bloater (<i>Coregonus hoyi</i>), alewife (<i>Alosa pseudoharengus</i>)
Metalimnion	Steelhead	Lake Michigan	TD	16	224	1.37	1.03	0.00	Mostly YOY bloater (<i>Coregonus hoyi</i>), alewife (<i>Alosa pseudoharengus</i>)
Metalimnion	Sturgeon	Lake Huron	TD	9	545	1.83	2.88	0.01	Rainbow smelt (<i>Osmerus mordax</i>)
Metalimnion	Sturgeon	Lake Michigan	TD	17	282	1	0.00	0.00	Mostly YOY bloater (<i>Coregonus hoyi</i>), alewife (<i>Alosa pseudoharengus</i>)
Metalimnion	Baird	Lake Huron	TS	7	115	1	0.59	0.00	Rainbow smelt
Metalimnion	Baird	Lake Michigan	TS	5	47	1	3.34	0.05	Mostly YOY bloater (<i>Coregonus hoyi</i>), alewife (<i>Alosa pseudoharengus</i>)
Metalimnion	Steelhead	Lake Michigan	TS	16	224	1	1.32	0.00	Mostly YOY bloater (<i>Coregonus hoyi</i>), alewife (<i>Alosa pseudoharengus</i>)
Metalimnion	Sturgeon	Lake Huron	TS	9	545	1.21	0.16	0.00	Rainbow smelt (<i>Osmerus mordax</i>)
Metalimnion	Sturgeon	Lake Michigan	TS	17	282	1	0.00	0.00	Mostly YOY bloater (<i>Coregonus hoyi</i>), alewife (<i>Alosa pseudoharengus</i>)
Hypolimnion	Baird	Lake Huron	NASC	4	127	1	5.47 *	0.03	YAO bloater (<i>Coregonus hoyi</i>), cisco (<i>Coregonus artedii</i>)
Hypolimnion	Baird	Lake Michigan	NASC	4	94	1	2.01	0.01	YAO bloater (<i>Coregonus hoyi</i>)
Hypolimnion	Steelhead	Lake Michigan	NASC	7	222	1.83	5.64 **	0.05	YAO bloater (<i>Coregonus hoyi</i>)
Hypolimnion	Sturgeon	Lake Huron	NASC	7	500	1.31	3.00	0.01	YAO bloater (<i>Coregonus hoyi</i>), cisco (<i>Coregonus artedii</i>)
Hypolimnion	Sturgeon	Lake Michigan	NASC	13	577	1	0.65	0.00	YAO bloater (<i>Coregonus hoyi</i>)
Hypolimnion	Baird	Lake Huron	TD	4	86	1.81	2.47	0.04	YAO bloater (<i>Coregonus hoyi</i>), cisco (<i>Coregonus artedii</i>)
Hypolimnion	Baird	Lake Michigan	TD	4	16	1	20.34 ***	0.56	YAO bloater (<i>Coregonus hoyi</i>)
Hypolimnion	Steelhead	Lake Michigan	TD	7	81	1	2.48	0.02	YAO bloater (<i>Coregonus hoyi</i>)
Hypolimnion	Sturgeon	Lake Huron	TD	7	41	1.67	1.89	0.04	YAO bloater (<i>Coregonus hoyi</i>), cisco (<i>Coregonus artedii</i>)
Hypolimnion	Sturgeon	Lake Michigan	TD	13	66	1.45	0.29	0.00	YAO bloater (<i>Coregonus hoyi</i>)
Hypolimnion	Baird	Lake Huron	TS	4	86	1	7.09 **	0.07	YAO bloater (<i>Coregonus hoyi</i>), cisco (<i>Coregonus artedii</i>)
Hypolimnion	Baird	Lake Michigan	TS	4	16	1	1.81	0.05	YAO bloater (<i>Coregonus hoyi</i>)
Hypolimnion	Steelhead	Lake Michigan	TS	7	81	1	0.31	0.00	YAO bloater (<i>Coregonus hoyi</i>)
Hypolimnion	Sturgeon	Lake Huron	TS	7	41	1	1.47	0.01	YAO bloater (<i>Coregonus hoyi</i>), cisco (<i>Coregonus artedii</i>)
Hypolimnion	Sturgeon	Lake Michigan	TS	13	66	1	0.30	-0.01	YAO bloater (<i>Coregonus hoyi</i>)

Note: * = significant at $\alpha = 0.05$, ** = significant at $\alpha = 0.01$, *** = significant at $\alpha = 0.001$; The Steelhead did not do overtakes in Lake Huron.

Note 2: The epilimnion was from 5 to 17 m, the metalimnion was from 18 to 40 m, and the hypolimnion was from 40 to 80 m.

[†]EDF = effective degrees of freedom. EDF values of 1 are the result of the GAM estimating a simple linear fit.

*Suspected fish species are the predominant species found by annual prey fish trawl surveys in these layers.

NASC = nautical area scattering coefficient, TD = target depth, TS = target strength.

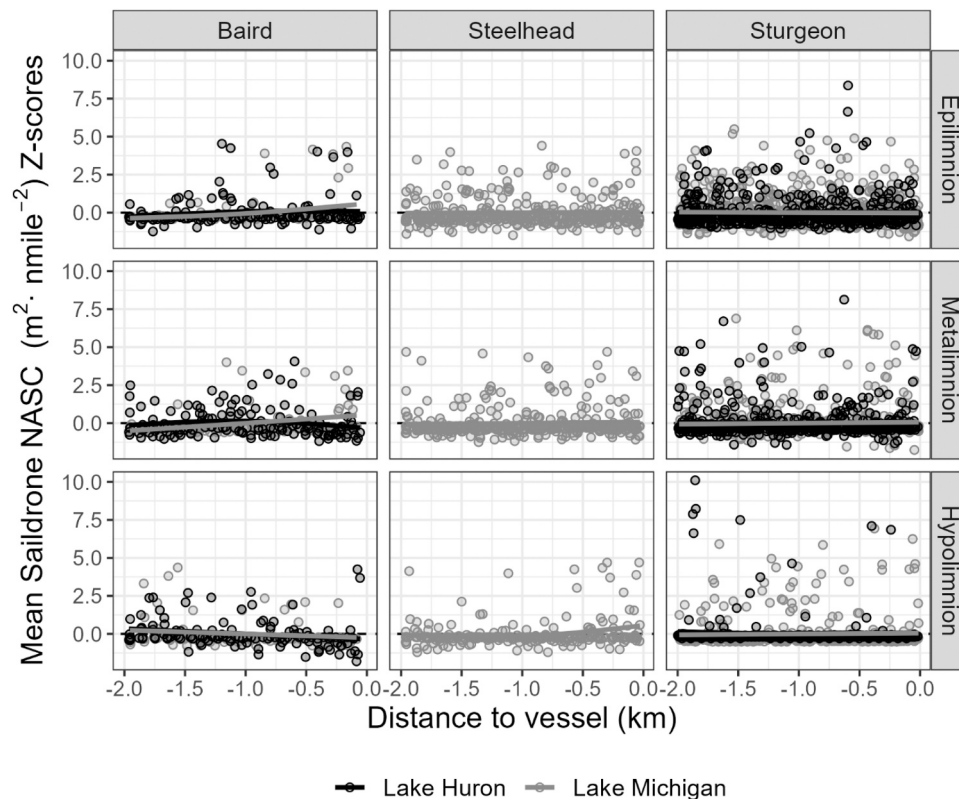


Fig. 4. Z-scores (calculated for each overtake) of Uncrewed Surface Vessels (USVs) Nautical Area Scattering Coefficient (NASC) values (integrated over 30 s intervals) regressed against the distance to the overtaking vessel (in km) with general additive models. Separate models were fit to each vessel, depth layer, and lake resulting in a total of 15 models.

not a systematic response of fish to vessels, may be a driving factor in the observed differences between USVs and motorized survey vessels. The Baird and Sturgeon are larger vessels (~30 m) than the Steelhead (19 m; Table 1). As a result, we anticipated the Sturgeon and the Baird, which radiate about the same noise level into the water column, would elicit stronger avoidance by fish, followed by the quieter Steelhead. In contrast to the survey vessels, USVs are small (7 m) compared to ships, have only small electric motors to control onboard systems, no propellers, and did not generate noise louder than the ambient and thus about 20–40 dB re 1 μ Pa (2–4 orders of magnitude) lower than the lowest noise ship (Steelhead). Therefore, it is reasonable to assume that fish would not respond to USVs (although this was not tested in the present study).

Target Strength may increase before declining again when fish dive as the swimbladder, the largest contributor to TS in a fish, can be angled in the body of the fish. This is the case for at least one of the fish species of interest here – alewife (*Alosa pseudoharengus*) – which has the highest TS when tilted 20 degrees down (i.e., diving) as the swimbladder becomes perpendicular to the transducer (Horne et al., 2000; Reeder et al., 2004; Rudstam et al., 2012). We did observe a case where TS and TD changed simultaneously (epilimnion of Baird in Michigan, hypolimnion of Baird in Huron), and this may suggest that diving was occurring. However, this did not contribute to differences in NASC along the transect.

Our results suggest limited effects of fish avoidance in Lakes Huron and Michigan with the current complement of survey vessels. The

strongest evidence for avoidance was with the Baird (Table 2), which was one of the loudest overall ships (Table 1); its overall noise levels were higher across almost all octaves (Fig. S5). It is surprising that we did not see a stronger response, as avoidance behavior was inferred in both Lake Ontario (Elliott et al., 2021) and Lake Superior (Grow et al., 2020) for surface-oriented fish (top 2–10 m) with different but similar vessels (≥ 20 m). However, fish avoidance did not occur in all lakes in a separate study which used a much smaller vessel (5.5 m) in multiple waterbodies (Wheeland and Rose, 2015), indicating context specific responses. Here, we did not find strong evidence that fish, primarily alewife, rainbow smelt (*Osmerus mordax*), and bloater (*Coregonus hoyi*), in Lakes Michigan and Huron responded to survey vessels although we note that these results are only valid for water depths below 5 m. This despite the noise levels from the three survey vessels in the present study exceeding the ICES recommendation for acoustic survey vessels at some frequencies. Prior studies have generally compared only two platforms to each other (e.g., Grow et al., 2020; Wheeland and Rose, 2015), often in scenarios where a single fish species predominates (e.g., De Robertis et al., 2008; Elliott et al., 2021); our work is novel in providing comparisons between multiple survey vessels in waters with mixed fish communities and at numerous locations.

While we found little evidence of fish avoidance, our study would have been unlikely to detect < 50% differences in acoustic measures given the sample sizes we were able to achieve and the sample variability we observed (Table S3). Fish densities, and hence acoustic return, are often low in the Great Lakes (Fig. S2), and small differences in

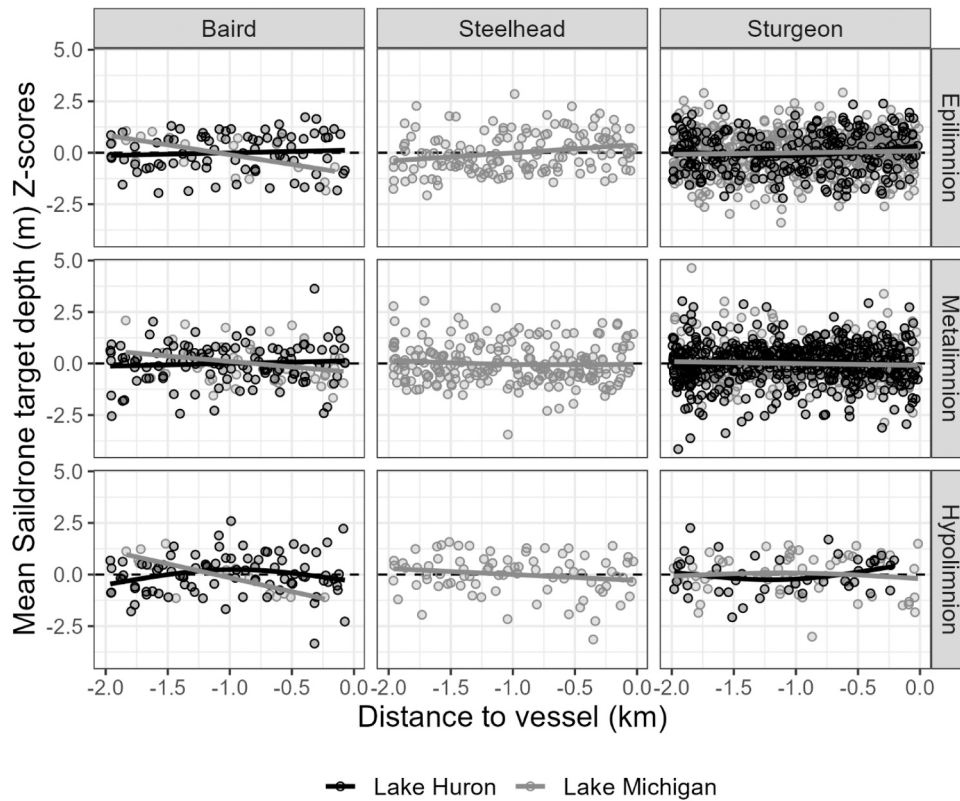


Fig. 5. Z-scores (calculated for each overtake) of Uncrewed Surface Vessels (USVs) target depth values (integrated over 30 s intervals) regressed against the distance to the overtaking vessel (in km) with general additive models. Separate models were fit to each vessel, depth layer, and lake, resulting in a total of 15 models.

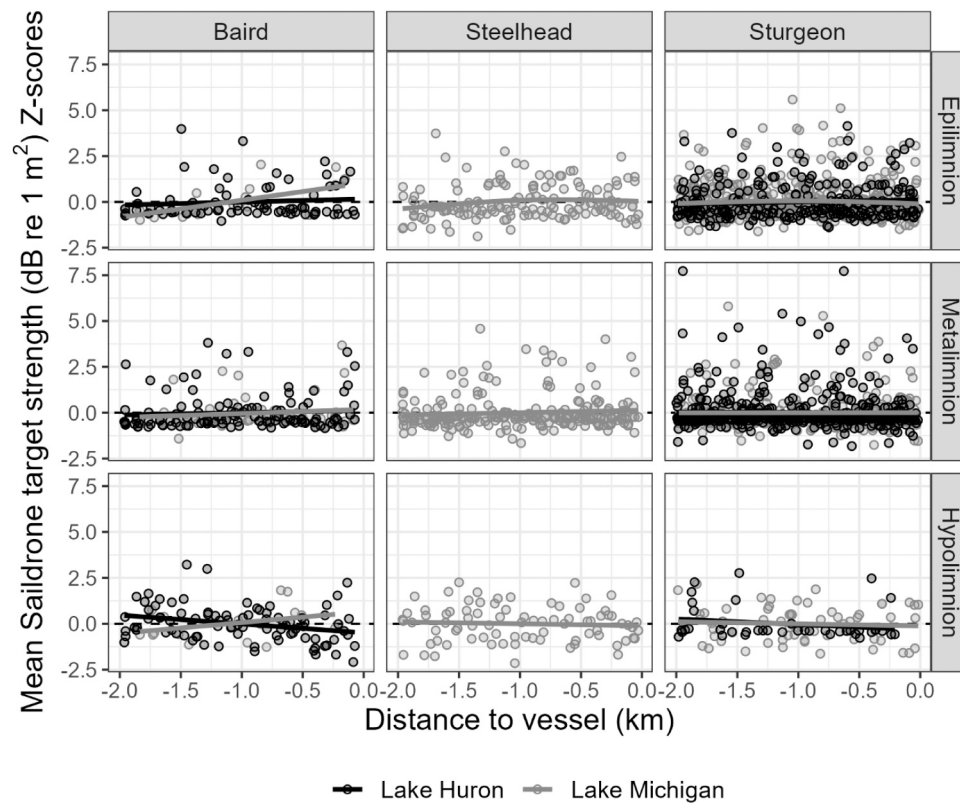


Fig. 6. Z-scores (calculated for each overtake) of Uncrewed Surface Vessels (USVs) target strength values (integrated over 30 s intervals) regressed against the distance to the overtaking vessel (in km) with general additive models. Separate models were fit to each vessel, depth layer, and lake, resulting in a total of 15 models.

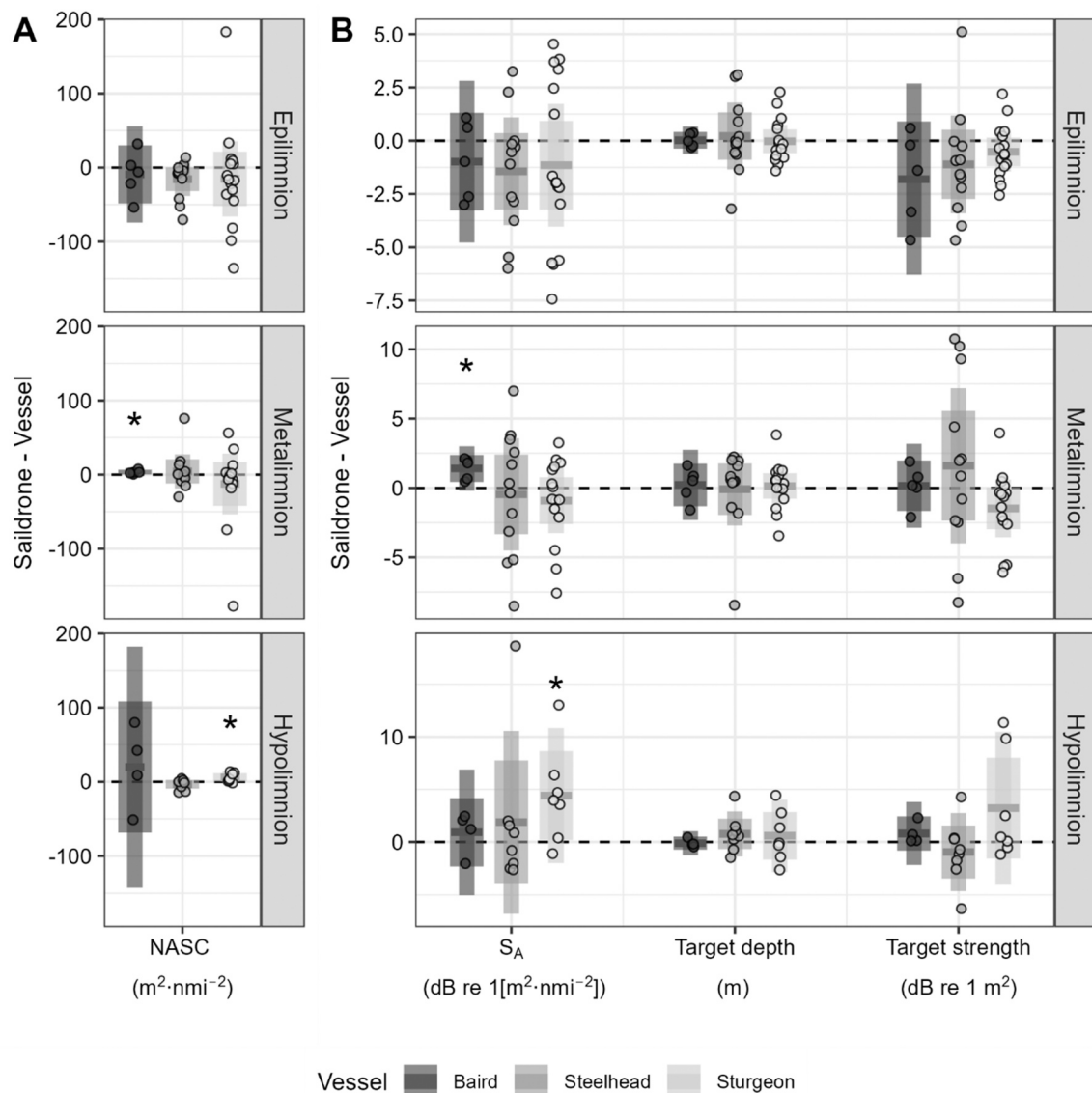


Fig. 7. Mean difference between the uncrewed surface vessel (USV) and overtaking vessel's (A) nautical area scattering coefficient (NASC) and (B) S_A , target depth and target strength for the 2 km parallel overtakes separated by depth layer. S_A is $10 \cdot \log_{10}[\text{NASC}]$ and the difference in S_A therefore is the logged ratio of drone and vessel NASC multiplied by ten. Points show each overtake, the 95% confidence intervals (CI) are represented by the shorter vertical bars, and 99% CI are the longer vertical bars. The dashed horizontal lines indicate no difference between platforms; values that fall below the dashed line favor the vessel, while those above favor the USV. If the 95% CI do not overlap zero a '*' appears above the bar, indicating a statistically significant difference between the USV and the respective survey vessel for a given acoustic metric.

acoustic response are likely to be obscured by variability. While we cannot rule out that some avoidance may occur, our overall finding that fish avoidance of survey vessels is small in magnitude if present, suggests preyfish surveys are not strongly affected by using multiple vessels for the Michigan and Huron surveys.

For fish behavior to bias acoustic survey data, fish must both detect approaching ships and have time to respond before the ships pass over them. Sound detection by fishes is species dependent but most can detect sound frequencies up to 1000 Hz with the lower frequencies detected by the lateral line (Mitson, 1995; Popper et al., 2022). Lower vessel-generated noise, especially for frequencies lower than 1 kHz is believed to decrease fish avoidance and lead to less biased acoustic surveys (Mitson, 1995). As a result, the ICES recommendations for "silent" ships are primarily to limit fish avoidance at lower frequencies based on expectations around fish sound detection and at high frequencies to prevent noise contamination in echosounders (Mitson, 1995). For this study, we found that the survey vessels likely exceeded

the ICES recommendations for underwater radiated noise at some frequencies, primarily in the 10–100 Hz frequency range (Figs. S5, S6). The vessel noise produced in this range can be primarily attributed to the propeller at the blade pass frequency and its harmonics. At frequencies above 1000 Hz, vessel generated noise decreased (Fig. 3).

A vessel's presence may have a larger effect on fish in the surface blind zone (in the current study <5 m deep for all vessels and the USV). The blind zone is caused by the transducer nearfield and because hull-mounted acoustic transducers (such as the ones used in this study) are submerged to some depth (Scalabrini et al., 2009; Totland et al., 2009). If the fish move down from the surface blind zone as vessels approach, the result may be a higher NASC in vessels than USVs. Prior work with upward looking transducers on a towed underwater vehicle in Lake Superior found 2.5 times more fish than a ship-based downward-aimed transducer, with differences primarily from the uppermost 14 m of the water column (Grow et al., 2020). The lack of information on surface-oriented fish is a problem with acoustic surveys and needs

further investigations. Unfortunately, the blind zone on the USVs is as large or larger than the survey vessels since the transducer is mounted at the bottom of the keel, 2 m depth. More work with upward-oriented transducers (for Great Lakes examples, Elliott et al., 2021; Grow et al., 2020; Riha et al., 2017) and multibeam sonar in horizontal beaming (e.g., omnidirectional; Vatnehol et al., 2018) is needed to better understand how the surface blind zone affects acoustic estimates.

The fish species being surveyed are likely to affect the results, making vessel avoidance context specific. For instance, clupeids are often better at detecting sound than many other fish groups (Popper et al., 2004). In Lake Huron there are fewer alewives (Hondorp et al., 2022) than Lake Michigan (Tingley et al., 2021) which may be the reason for a tendency for more avoidance behavior in Lake Michigan than in Lake Huron (Figs. 4–6). In Lake Ontario, where alewife predominate, a stationary upwards looking hydroacoustic unit observed larger and more numerous targets than when the R/V Kaho conducted active acoustic surveys (Elliott et al., 2021). This could be due to fish present in the top 5 m of the water column, which were not included in down-looking data collected by the R/V Kaho, or indicate that avoidance would be more important in lakes dominated by alewife (see also Riha et al., 2017).

Uncrewed surface vessels are a relatively new technology that show promise for evaluating fish avoidance in acoustic survey vessels. To our knowledge this study presents the first test of fish responses to motorized survey vessels with USVs in freshwaters although investigations into fish avoidance in marine environments using these USVs are available from a small number of recent studies (Chu et al., 2019; De Robertis et al., 2019). In the most complete comparison of acoustic data to date between ship and USV of the type used in this study, De Robertis et al. (2019) found evidence in marine environments that this type of USV detected more walleye pollock (*Gadus chalcogrammus*) at shallower depths but detected fewer fish in the deepest water compared to the large (64 m) motorized survey vessel R/V Oscar Dyson. This pattern persisted across both day and nighttime sampling, but during daytime pollock appeared to dive, while during night there was evidence for lateral avoidance or changes in their orientation as the vessel approached (De Robertis et al., 2019). In a more limited study off Vancouver Island with a sister ship, a USV of the type used in this study was overtaken by a vessel once and observed to have higher NASC values than the trailing vessel (Chu et al., 2019). In contrast to these prior studies, our parallel overtakes found little difference in NASC, average target depth or TS values (Fig. 7). Our vessels were louder and slower (Oscar Dyson class ships survey at $\sim 5.5 \text{ m}\cdot\text{s}^{-1}$) (De Robertis et al., 2008; Stimpert et al., 2019), but considerably smaller (less than half the length), than vessels used in prior comparison studies with these USVs; thus the relative lack of fish responses detected in our study may indicate that the physical size of the vessel and the water displacement are also important component of fish avoidance, and not only vessel generated noise.

Use of long-range USVs capable of continuous acoustic measurements is increasing (De Robertis et al., 2021; Verfuss et al., 2019). Uncrewed Surface Vessels operating without engines and therefore being effectively silent (Fig. S5), are capable of providing an important contrast to ship-based acoustic surveys and are well-suited to exploring questions around potential bias in fisheries acoustic surveys. We found little evidence that fish avoided the vessels measured in this study even though they operate at noise levels above ICES recommendations for noise reduction thresholds. Understanding ship avoidance remains an important challenge for acoustic surveys but results from these trials in the Laurentian Great Lakes indicate that fish behavioral responses do not appear to be a simple function of vessel noise. Further work on understanding fish behavioral responses to any fishing gear, including acoustics, could improve our ability to assess fish populations and by extension our ability to apply ecosystem-based fisheries management to lakes and seas (Link, 2017).

CRediT authorship contribution statement

Thomas M Evans: Data curation, analysis, writing, editing. **Lars G. Rudstam:** Funding acquisition, methodology, writing, editing, supervising. **Suresh A. Sethi:** Funding acquisition, methodology, writing, editing, supervising. **David M Warner:** Methodology, field work, editing. **S. Dale Hanson:** Methodology, field work, editing. **Benjamin Turschak:** Methodology, field work, editing. **Steven A. Farha:** Methodology, field work, editing. **Andrew Barnard:** Funding acquisition, methodology, field work, writing, editing, supervising. **Daniel L. Yule:** Methodology, editing. **Mark R. Dufor:** Methodology, field work, editing. **Timthoy P. O'Brien:** methodology, field work, editing. **Kevin N. McDonnell:** Methodology, editing. **James M. Watkins:** Funding acquisition, methodology, editing. **Scott R. Koproski:** Methodology, field work, editing. **Susan E. Wells:** Methodology, editing. **Patricia M Dieter:** Methodology, field work, editing. **Erik Kocher:** Methodology, field work. **James J. Roberts:** Methodology, editing. **Steven Senczyn:** Field testing, data analysis, writing, editing. **Peter C. Esselman:** Project administration, methodology, field work, editing.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Thomas Evans reports financial support was provided by US Geological Survey.

Data Availability

Data are available at: <https://doi.org/10.5066/P9EP1XIG>.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.fishres.2023.106817](https://doi.org/10.1016/j.fishres.2023.106817).

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